

The 5-minute demo, what to say, and how every part works

Read this out loud three times before judging. One person speaks, the other drives the screen. Every number below was measured by running our code.

0 · The one rule for this demo: proud and precise

Judges at this level ask “is this real?”. The strongest answer is a precise one. We have **working software, firmware and a trained model**. Some links are simulated because we own only one relay. Saying that clearly, before they ask, earns more trust than any claim. Use these exact labels:

Part	Say it like this
Phone → SOS page → dashboard → Read/Dispatch back to phone	WORKING “This is live. Send one yourself.”
AI semantic compression, byte and airtime numbers	WORKING “Trained model, real numbers for the SOS you just sent.”
QAOA relay planner in Qiskit	WORKING “Real Qiskit circuit, run on the Aer simulator, checked against brute force.”
Rescuer node + gateway LoRa link	If the boards are on the table: LIVE . If not: “Our teammate is field-testing the radios. Today the link runs over USB with the same messages and the same retry/ACK rules.”
Multi-hop relays, self-healing	SIMULATED “We have one relay, so the multi-hop rules run in a simulator. It is the same code the firmware runs.”
Survivor-density heatmap	SCENARIO “Estimated density for this disaster scenario. With the rescuer node live, it comes from real phone counts.”
Sending records to SACHET / 108 / hospitals	PROPOSED “Designed and the export works; not connected to government systems yet.”

1 · Demo script (5 minutes)

0:00 · HOOK (SPEAKER)

“Earthquake in Musheerabad, Hyderabad. The towers are down. You're trapped on the second floor, your phone says *No service*, and nobody knows you're there. Today you're the survivor.” (*Hand the judge the phone.*)

0:20 • PROBLEM

“Phones survive disasters; networks don't. Government alerts are one-way. Satellite SOS needs new phones and open sky. Mesh radios need everyone to own a device beforehand. And even when an SOS gets out, the person never learns if anyone read it.”

0:40 • WHAT WE BUILT

“VOID-NAV is a private network that rescuers carry in. A rescuer node opens a Wi-Fi called SOS Node1. Any phone can join it, with no app and no SIM. It sends the SOS over long-range LoRa radio to the command post, and the answer comes back.”

DRIVER

Phone on **SOS Node1**, or the laptop page if on emulated mode. Dashboard on **SOS Inbox**, alarm on.

1:00 • JUDGE SENDS AN SOS

“Tap what happened, set the people, and type anything, in any language.” (*Judge sends.*) “It shows Sending, then Delivered. The node retries until the command post confirms.”

1:20 • IT LANDS

(*Siren + voice.*) “Critical first. Your own words, decoded into a triage card: trapped, bleeding, three to five people, needs water.”

1:40 • SEMANTIC COMPRESSION

(*Click ✂ Semantic compression demo.*) “Long-range radio is slow: about 1.8 kilobits per second. So we don't send your words; we send their meaning. Our AI reads your message and packs it into five bytes. Your message was 142 bytes on air; this is 15. That's 78% less airtime, so four to five times more SOS fit through the same channel.”

2:20 • READ & DISPATCH

(*Open the card → phone beeps “Read”. Press Dispatch / 108 ambulance → phone shows “Help is on the way”. Send reply.*) “Opening it tells you a human saw it. Dispatch tells you help is coming. That's the loop nobody else closes.”

2:50 • MAP

“The map shows where survivors are concentrated, which tells rescuers where to search first. Tap a hospital to see its real phone numbers and distance.”

3:10 • QUANTUM

(*Relays · QAOA → Load rooftops → Import SOS + heatmap → Run in Qiskit Aer.*) “A rescue team has only a few relays. Which rooftops cover the most survivors, and how many relays do we need? That's a combinatorial problem, and QAOA is the quantum algorithm built for it. One qubit per rooftop. Qiskit says four relays are needed, and brute force confirms it's the best possible plan.”

3:50 • SELF-HEALING

(*Continue → Self-healing → Run disaster scenario.*) “Now an aftershock destroys a relay. Nobody tells the network what to do. Each node notices a missing acknowledgement and reroutes on its own. When the survivor is completely cut off, messages are stored, not lost. When a relay comes back, they flush through. Forty out of forty delivered; a fixed route delivers six.”

“Built on ₹1,200 nodes, with no internet anywhere. An instant private network rescuers carry in. Any phone becomes a beacon, and it gets an answer back.”

2 · How the rescuer node finds phones and sends the SOS

Finding phones (the heatmap)

Every smartphone with Wi-Fi switched on shouts “is my home network here?” every few seconds, even when it isn't connected. These are called **probe requests**. The ESP32 can listen for them (**promiscuous mode**) while it runs its hotspot.

- Every 20 s it **counts distinct phones** and the strongest signal (RSSI, in dBm: -40 is very close, -85 is far).
- It sends only “**6 phones, strongest -52 dBm**” over LoRa. Phone addresses are scrambled (hashed) in memory for 20 s and never stored or sent.
- The dashboard paints each count where the rescuer is standing. Walking the site builds the heatmap: red means many phones, so search there first.
- **Limits to state honestly:** phones randomise their Wi-Fi address, so counts are approximate. A phone with Wi-Fi off or a dead battery can't be found. A dog, radar or thermal camera finds those people.

Connecting and sending

1. The rescuer shouts “join Wi-Fi SOS Node1”. It's an open network at the top of the list. The phone joins and a **captive portal** opens the SOS page automatically, the same way hotel Wi-Fi does. The page is at 192.168.4.1.
2. The survivor taps answers and types a note. The page sends it to the ESP32.
3. The node turns it into a compact packet (SemCode) and transmits it over **LoRa 433 MHz, SF9**. It **retries every ~3 s, up to 8 times**, until the command post sends an ACK.
4. The gateway ESP32 at the command post receives it and passes it to the laptop over USB. The dashboard shows it, and every status (READ / DISPATCH / REPLY) goes back the same way. The phone page checks every second and beeps on each change.

3 · AI semantic compression, explained simply

The problem: LoRa reaches kilometres, but it is slow and the channel is shared. Long text messages take long to send, collide with each other and drain batteries.

The idea (semantic communication, a 6G research direction): for an SOS, rescuers need a handful of facts, not the exact wording. Send the *meaning*.

Step	What happens	Words to use
1	The survivor types anything in English, Hindi, Telugu or Hinglish.	“free text, multilingual”
2	Our model reads it and predicts the triage fields: category, injured, bleeding, unconscious, people, position, vulnerable people, needs.	“text classification”, “17 prediction heads”
3	A shared codebook packs the fields into 40 bits = 5 bytes , e.g. category = 4 bits, people = 3 bits, needs = 8 bits.	“SemCode token”, “bit-packing”
4	5 bytes + a 10-byte header go over the radio. Airtime is computed with the official LoRa formula.	“time-on-air, Semtech AN1200.13”
5	The dashboard expands the 5 bytes back into a full sentence using the same codebook.	“decoder”

The model: a **Naive Bayes ensemble** (one model per field), trained on **4,200** SOS messages generated in 5 language styles. Its features are words, word pairs and a multilingual root-word list (*phas, chikk, stuck* → trapped; *khoon, rakt, bleed* → bleeding).

Accuracy on wording it never saw: category 83%, bleeding 97%, injured 93%. On 30 hand-written messages, category is right 90% of the time.

Why not ChatGPT-style AI? It's too big for a node, needs internet, and can make details up. Ours is tiny, offline and deterministic. **Taps always win:** the AI can only add information or raise urgency, never lower it.

Honest note: training data is synthetic, because real SOS logs aren't public. The next step is to retrain on real drill data.

4 · QAOA and Qiskit, explained so anyone gets it

The question

“We have 8 possible rooftops and only a few relays. Where do we put them to reach the most survivors, and what's the fewest relays that reaches 90%?” Choosing K rooftops out of N is a **combinatorial optimisation** problem. The number of options explodes: 8 rooftops is 56 choices, but 100 rooftops with 10 relays is 17 trillion.

The analogy

Think of a quantum computer as trying **all plans at once** as a blurred “maybe” (**superposition**). QAOA is a recipe to **tilt the odds** so that, when you finally look (**measure**), good plans come up much more often than bad ones.

Technical term	Plain meaning
Qubit	One per rooftop: 1 = relay here, 0 = no relay.
Cost Hamiltonian	A score for each plan: + people covered, – people counted twice. Built into the circuit as RZ and RZZ gates.
Dicke state	Start with an equal mix of every plan that uses exactly K relays.
XY mixer (RXX + RYY gates)	Moves a relay from one rooftop to a neighbour, but never adds or removes one, so every answer is valid. Textbook QAOA wastes shots on invalid plans.
Layers (p)	Repeat “score, then shuffle” p times (we use 2). More layers give better odds.
γ, β (angles)	Two dials per layer: how hard to push toward good plans, and how much to shuffle.
Hybrid loop / COBYLA	A normal computer turns the dials; the quantum circuit says how good the result is. Repeat until it stops improving.
Shots	Run the final circuit 8,192 times and count which plans come out. Keep the best.
Brute force check	We also try every plan the slow way to prove QAOA's answer is the true best.

How we ran it in Qiskit (say this when asked)

“We wrote the circuit in **Qiskit 2.5**. `initialize` creates the Dicke state, `rz/rzz` form the cost layer, and `rx+ry` form the mixer ring. We optimised the angles with **SciPy COBYLA** on Qiskit's Statevector, then transpiled the circuit and ran **8,192 shots on the AerSimulator**. The dashboard sends the rescuer's map inputs to that script.”

Result on the 8-site map: **4 relays needed**, covering 59 of 64 people. Brute force agrees at every K. Every shot placed exactly K relays. For K = 3, QAOA found the best plan 16% of the time versus 4% for a random guess.

Say before they ask: “At 8 rooftops brute force is faster, and we don't claim a speed-up. We claim a correct quantum formulation, verified against the exact answer, that grows to sizes brute force can't handle, and can run on IBM quantum hardware with the same code.”

5 • Self-healing with only one relay

We own one relay, so we show multi-hop healing in a simulator that uses the same rules as the firmware. The dashboard draws it on the real map with the relays QAOA chose.

1. **Beacon:** every 30 s, each node says “I am N hops from the gateway”.
2. **Neighbour table:** a neighbour silent for 90 s is dropped.
3. **Gradient:** my hops = best neighbour + 1. The network redraws its own map after any loss.
4. **Closer-only forwarding:** pass the SOS only to a neighbour closer to the gateway, which must ACK.
5. **No ACK in 2 s:** try the next-best neighbour. This is the healing.
6. **Store and carry:** no route at all → keep the SOS until a route returns. Nothing is dropped.

Scenario result: **40/40 delivered with healing vs 6/40 on a fixed route**; zero SOS lost while cut off. Team simulator: 119/119 vs 29/119.

With our real relay: one hop is real (node → relay → gateway); the chain of many relays is the simulated part.

6 · Is the dashboard offline? Which parts need internet?

Part	Internet?
Survivor SOS page, rescuer node, LoRa, gateway	Never. All local radio.
Command server, inbox, alerts, voice, AI model, compression demo, QAOA (browser), self-healing	No. Runs on the laptop; fonts, map library and ward boundaries are bundled.
Street-map background tiles	Yes , when online. Offline the map still shows real ward boundaries, heatmap, SOS and hospitals. Production: pre-download (cache) tiles for the district.
Qiskit button	No at run time. It needs <code>pip install qiskit qiskit-aer</code> once beforehand.
Sending records to SACHET / state control room / 108	Only if any backhaul exists: a satellite terminal, surviving fibre, or restored cellular. PROPOSED

7 · Do we need hospitals? How does data reach them?

No. The core loop works without any hospital connection. Hospitals and ambulances benefit in three ways:

1. **Phone contacts on the map:** tap a hospital to get its published numbers and distance, for the commander to call (satellite or restored phone).
2. **Viewer role on the laptop's Wi-Fi:** an ambulance crew or field hospital nearby opens `/command?viewer=1` on any phone and sees the live, read-only inbox.
3. **CAP export** **PROPOSED** : each SOS can be downloaded as a **CAP 1.2** record (Common Alerting Protocol, the format India's SACHET uses). When backhaul exists, the laptop forwards these to state control rooms and the 108 service.

8 · Judge Q&A, short answers

No phone, or phone dead?

Nothing can transmit, and no network can hear it. The node's push button sends a one-press SOS for anyone a rescuer finds. Dogs, radar and thermal cameras find the rest, and their findings travel over our network.

How do survivors know to join?

Rescuers shout it, it's the open network at the top of the list, and it can go in pre-disaster drills and the official alert SMS.

Range?

Phone to node: tens of metres of Wi-Fi, which is why rescuers carry the node in. Node to node: typically 1–5 km on LoRa in cities (published typical values, not yet measured by us).

Is 433 MHz legal?

The demo runs at low power. Production moves to India's 865–867 MHz licence-free band; it's a one-line firmware change.

Security / fake SOS?

The prototype is unencrypted. Roadmap: a per-deployment AES key plus an authentication tag on every packet. No names or numbers are collected.

How is this different from Meshtastic / Project OWL?

Any phone with no app; a closed Read/Dispatch loop to the survivor; AI compression; relay planning; and a working command dashboard.

Why quantum at all?

The radios are classical and must be. Quantum belongs where the hard maths is: planning relays at city scale. We show a correct, verified formulation today.

Is the AI really trained?

Yes. The training script and data generator are in the repo, it retrains in 3 seconds, and the accuracy is measured on wording it never saw.